AIMS
Airbus Material Specification

Aluminium Alloy (7040/7050)
Solution treated, controlled stretched and artificially aged (T7451)
Plate
75,0 mm ≤ a ≤ 220,0 mm
High toughness
Close tolerance flatness

Material Specification

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Contents list

1 Scope

This Material Specification defines the requirements of aluminium alloy 7040/7050 T7451 plate 75,0 mm ≤ a ≤ 220,0 mm for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

No typical applications.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 03-02-000	Airbus Industrie Material Specification. Technical Specification – Aluminium and aluminium alloy wrougth products – plate.
EN 2032-2	Aerospace series - Metallic materials - Part 2: Coding of the metallurgical condition in delivery condition.
EN 4500-2	Aerospace series – Instructions for the drafting and use of metallic material standards. Part 2: Coding of the metallurgical condition in delivery condition.
EN 6072	Aerospace series - Metallic materials - Test Method - Constant amplitude fatigue testing.
ASTM G34	Standard Test Method for Exfoliation Corrosion Susceptibility in 2xxx and 7xxx Series Aluminium Alloys (EXCO Test)
ABS 5324	Aerospace series - Plates - Aluminium alloy 7040/7050 – High Toughness - Close tolerance flatness - Thickness 75 mm ≤ a ≤ 220 mm - Dimensions

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.2 Specific

See tables according to EN 4500-2.

5 Designation

See ABS 5324

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 03-02-000.

1	Material designation		Aluminium alloy 7040/7050										
2	Chemical Composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Others Each	Others total	Al
		min.	See relevant IPS										
		max.											
3	Method of melting		-										
4	Form Method of production Limit dimension (mm)		Plate Rolled $75 \leq a \leq 220$										
5	Technical specification		AIMS 03-02-000										
6	6.1 Delivery condition and heat treatment		T7451										
	6.2 Delivery condition code		U [1)]										
7	Use condition and heat treatment		T7451 As delivered										
8	Sample Test piece Heat treatment		T 7451										

9	Dimension concerned		a	mm	$75 \leq a \leq 100$			$100 < a \leq 125$			$125 < a \leq 150$		
10	Thickness of cladding on each face			%	-								
11	Direction of test piece				L	LT	ST	L	LT	ST	L	LT	ST
12	TENSILE	Temperature	θ	°C	Room temperature								
13		Proof stress	$R_{p0,2}$	MPa	430	430	410	425	425	405	425	420	400
14		Strength	Rm	MPa	495	495	475	490	490	470	485	485	470
15		Elongation	A	%	9	6	3	9	5	3	8	5	3
16		Reduction of area	Z	%	-								
17	Hardness		HB	-	-								
18	Shear strength		Rc	MPa	-								
19	Bending		k	-	-								
20	Impact strength		k	-	-								
21	CREEP	Temperature	θ	°C	-								
22		Time		h	-								
23		Stress	σ_a	MPa	-								
24		Elongation	a	%	-								
25		Rupture stress	σ_R	MPa	-								
26		Elongation at rupture	A	%	-								
27	Notes (see Line 98)				1) and 2)								

1	Material designation		Aluminium alloy 7040/7050

2	Chemical Composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Others		Al
											Each	total	
		min.	See relevant IPS										
		max.											

3	Method of melting				-								
4	Form Method of production Limit dimension (mm)				Plate Rolled 75 ≤ a ≤ 220								
5	Technical specification				AIMS 03-02-000								
6	6.1 Delivery condition and heat treatment				T7451								
	6.2 Delivery condition code				U [1]								
7	Use condition and heat treatment				T7451 As delivered								
8	Sample Test piece Heat treatment				T 7451								
9	Dimension concerned		a	mm	150 ≤ a ≤ 180			180 < a ≤ 200			200 < a ≤ 220		
10	Thickness of cladding on each face			%	-								
11	Direction of test piece				L	LT	ST	L	LT	ST	L	LT	ST
12	TENSILE	Temperature	θ	°C	Room temperature								
13		Proof stress	$R_{p0,2}$	MPa	425	415	395	420	415	390	420	405	385
14		Strength	Rm	MPa	475	475	455	470	470	455	470	470	455
15		Elongation	A	%	7	4	3	6	4	3	6	3	3
16		Reduction of area	Z	%	-								
17	Hardness		HB	-	-								
18	Shear strength		Rc	MPa	-								
19	Bending		k	-	-								
20	Impact strength		k	-									
21	CREEP	Temperature	θ	°C	-								
22		Time		h	-								
23		Stress	σ_a	MPa	-								
24		Elongation	a	%	-								
25		Rupture stress	σ_R	MPa	-								
26		Elongation at rupture	A	%	-								
27	Notes (see Line 98)				1) and 2)								

<table>
<tr><td rowspan="3">32</td><td rowspan="3">Electrical conductivity</td><td rowspan="3">7</td><td colspan="3">$\gamma \geq 23{,}0$ MS/m</td><td colspan="2">Acceptable</td></tr>
<tr><td colspan="3">22,0 MS/m $\leq \gamma <$ 23,0 MS/m</td><td colspan="2">Acceptable if $R_{p0,2}$ min LT + 70 MPa or stress corrosion test is acceptable.</td></tr>
<tr><td colspan="3">$\gamma <$ 22,0 MS/m</td><td colspan="2">Not acceptable</td></tr>
<tr><td rowspan="5">39</td><td rowspan="5">Stress corrosion</td><td rowspan="4">6</td><td colspan="5">ST-direction, constant load, tension</td></tr>
<tr><td colspan="5">Length of exposure: 30 days</td></tr>
<tr><td colspan="3">Dimensions
mm</td><td colspan="2">Stress level
MPa</td></tr>
<tr><td colspan="3">$75 \leq a \leq 220$</td><td colspan="2">240</td></tr>
<tr><td>7</td><td colspan="5">Criteria : no failure</td></tr>
<tr><td rowspan="7">40</td><td rowspan="7">Fracture toughness (K_{IC})</td><td rowspan="7">7</td><td>Dimension, mm</td><td>L-T
MPa√m</td><td colspan="2">T-L
MPa√m</td><td>S-L
MPa√m</td></tr>
<tr><td>75≤a≤100</td><td>34</td><td colspan="2">29</td><td>27</td></tr>
<tr><td>100<a≤125</td><td>33</td><td colspan="2">27</td><td>26</td></tr>
<tr><td>125<a≤150</td><td>32</td><td colspan="2">25</td><td>26</td></tr>
<tr><td>150<a≤180</td><td>30</td><td colspan="2">24</td><td>25</td></tr>
<tr><td>180<a≤200</td><td>29</td><td colspan="2">24</td><td>25</td></tr>
<tr><td>200<a≤220</td><td>28</td><td colspan="2">24</td><td>24</td></tr>
<tr><td>44</td><td>External defects</td><td>-</td><td colspan="5">See AIMS 03-02-000</td></tr>
<tr><td rowspan="20">46</td><td rowspan="20">Fatigue</td><td>3</td><td colspan="5">T-Type specimen in accordance with EN 6072,
$75 \leq a \leq 150$ mm</td></tr>
<tr><td rowspan="7">7</td><td colspan="5">The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by following data:</td></tr>
<tr><td>$\sigma_{max\ net}$ in MPa</td><td>Number of Cycles</td><td>R</td><td>Kt</td><td>Direction</td></tr>
<tr><td>290</td><td>1E04</td><td rowspan="5">0,1</td><td rowspan="5">2,3</td><td rowspan="5">T-L</td></tr>
<tr><td>210</td><td>3E04</td></tr>
<tr><td>155</td><td>1E05</td></tr>
<tr><td>145</td><td>2E05</td></tr>
<tr><td>130</td><td>1E06</td></tr>
<tr><td>3</td><td colspan="5">T-Type specimens in accordance with EN 6072,
150<a≤220 mm</td></tr>
<tr><td rowspan="8">7</td><td colspan="5">The curve of the test results, calculated in accordance with EN6072, shall not lie below the reference curve defined by following data:</td></tr>
<tr><td>$\sigma_{max\ net}$ in MPa</td><td>Number of Cycles</td><td>R</td><td>Kt</td><td>Direction</td></tr>
<tr><td>260</td><td>1E04</td><td rowspan="6">0,1</td><td rowspan="6">2,3</td><td rowspan="6">T-L</td></tr>
<tr><td>210</td><td>2E04</td></tr>
<tr><td>185</td><td>3E04</td></tr>
<tr><td>140</td><td>1E05</td></tr>
<tr><td>120</td><td>2E05</td></tr>
<tr><td>110</td><td>1E06</td></tr>
</table>

<table>
<tr><td>46</td><td>Fatigue</td><td>3</td><td>Mini T-Type specimen in accordance with EN 6072,
150 < a ≤ 220 mm</td></tr>
<tr><td></td><td></td><td>7</td><td>The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by following data:
<table>
<tr><td>$\sigma_{max\ net}$ in MPa</td><td>Number of Cycles</td><td>R</td><td>Kt</td><td>Direction</td></tr>
<tr><td>180</td><td>2E04</td><td rowspan="5">0,1</td><td rowspan="5">2,3</td><td rowspan="5">S-L</td></tr>
<tr><td>140</td><td>4E04</td></tr>
<tr><td>120</td><td>6E04</td></tr>
<tr><td>100</td><td>1E05</td></tr>
<tr><td>90</td><td>1E06</td></tr>
</table></td></tr>
<tr><td>46</td><td>Fatigue (Smooth Fatigue Test)</td><td>-</td><td>See AIMS 03-02-000</td></tr>
<tr><td></td><td></td><td>6</td><td>Thickness: 150 mm ≤ a ≤ 220 mm.</td></tr>
<tr><td></td><td></td><td>7</td><td>The specimens shall be tested a R= 0.1, at a maximum stress of 241MPa and shall meet the following fatigue requirements:
- Minimum individual life time: 90000 cycles.
- Log average of 4 samples: 120000 cycles.
- Runout 200000 cycles.</td></tr>
<tr><td>49</td><td>Exfoliation corrosion</td><td>7</td><td>Equal or better grade EB of ASTM G34 [2)]</td></tr>
<tr><td>61</td><td>Internal defects</td><td>-</td><td>See AIMS 03-02-000</td></tr>
<tr><td>69</td><td>Elastic modulus</td><td>7</td><td>The value must be within the following range
68000 ≤ E ≤ 72000 MPa
71000 ≤ E_c ≤ 75000 MPa</td></tr>
<tr><td>70</td><td>Compression Proof stress
$R_{0,2C}$</td><td>7</td><td>
<table>
<tr><td>Dimension
mm</td><td>L
MPa</td><td>LT
MPa</td><td>ST
MPa</td></tr>
<tr><td>75 ≤ a ≤ 100</td><td>420</td><td>455</td><td>455</td></tr>
<tr><td>100 < a ≤ 125</td><td>415</td><td>450</td><td>450</td></tr>
<tr><td>125 < a ≤ 150</td><td>410</td><td>440</td><td>440</td></tr>
<tr><td>150 < a ≤ 180</td><td>405</td><td>435</td><td>435</td></tr>
<tr><td>180 < a ≤ 200</td><td>400</td><td>430</td><td>430</td></tr>
<tr><td>200 < a ≤ 220</td><td>395</td><td>425</td><td>425</td></tr>
</table></td></tr>
<tr><td>71</td><td>Crack propagation</td><td>3</td><td>CCT specimen, W= 160 mm, 2 a_i = 4,0 mm
2 a_f = 53 mm, thickness = 5,0 mm, L-T and T-L</td></tr>
<tr><td></td><td></td><td>7</td><td>The crack propagation rates should not be greater than the following data:
<table>
<tr><td>ΔK in MPa√m</td><td>da/dN (mm/cycle)</td><td>R</td></tr>
<tr><td>5</td><td>2,0E-5</td><td rowspan="5">0,1</td></tr>
<tr><td>10</td><td>2,0E-4</td></tr>
<tr><td>20</td><td>2,0E-3</td></tr>
<tr><td>25</td><td>1,5E-2</td></tr>
<tr><td>30</td><td>5,0E-2</td></tr>
</table></td></tr>
</table>

<table>
<tr><td rowspan="6">82</td><td rowspan="6">Batch uniformity</td><td>-</td><td colspan="9">See AIMS 03-02-000</td></tr>
<tr><td rowspan="5">7</td><td colspan="3" rowspan="2">Electrical conductivity</td><td colspan="6">γ= 24,0 MS/m (typical value)</td></tr>
<tr><td colspan="3">$\delta \leq$ 1,0 MS/m per product</td><td colspan="3">$\Delta \leq$ 1,5 MS/m per batch</td></tr>
<tr><td colspan="9">or</td></tr>
<tr><td colspan="3" rowspan="2">Hardness</td><td colspan="6">140 HB (typical value)</td></tr>
<tr><td colspan="3">$\delta \leq$20 HB per product</td><td colspan="3">$\Delta \leq$ 30 HB per batch</td></tr>
<tr><td rowspan="5">85</td><td rowspan="5">Bearing stress
F_{bru}</td><td rowspan="5">7</td><td colspan="3">Dimensions
mm</td><td>75≤ a ≤100</td><td>100< a ≤125</td><td>125< a ≤150</td><td>150< a ≤180</td><td>180< a ≤200</td><td>200< a ≤220</td></tr>
<tr><td rowspan="2">Fbru</td><td rowspan="2">MPa</td><td>e/D= 2,0</td><td>1010</td><td>1000</td><td>980</td><td>960</td><td>950</td><td>930</td></tr>
<tr><td>e/D= 1,5</td><td>780</td><td>770</td><td>760</td><td>750</td><td>740</td><td>730</td></tr>
<tr><td rowspan="2">Fbry</td><td rowspan="2">MPa</td><td>e/D= 2,0</td><td>780</td><td>770</td><td>760</td><td>750</td><td>740</td><td>730</td></tr>
<tr><td>e/D= 1,5</td><td>650</td><td>640</td><td>630</td><td>620</td><td>605</td><td>590</td></tr>
<tr><td>98</td><td>Notes</td><td>-</td><td colspan="9">1) see EN 2032-2
2) Limitation: Test solution quantity 15 ± 1 ml / cm^2</td></tr>
<tr><td>99</td><td>Typical use</td><td>-</td><td colspan="9">- - - -</td></tr>
<tr><td>100</td><td>Qualification</td><td></td><td colspan="9">Frequency of qualification testing is included in AIMS 03-02-000
Recommended thickness are 100 mm, 150 mm, 170 mm, 190 mm and 215 mm</td></tr>
</table>

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 October 2000	--	New Standard
2 Sept 2001	Line 46 Line 71	Fatigue an crack propagation have been modified
3 October 2001	Line 71 Line 85	Typing mistakes correction
4 May 2002	Line 46 Line 71	Fatigue an crack propagation have been modified
5 July 2004	Line 2 Line 32 Line 71	Chemical composition, Electrical conductivity and Crack propagation have been modified
6 June 2005	Line 46	Fatigue Smooth Test requirements have been added.
7 January 2006	Line 82	Typing mistakes correction